EDUARDUS TJITRAHARDJA (徐和平)

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EDUCATION

TSINGHUA UNIVERSITY Beijing, China
Master's in Advanced Computing *Sep 2025 – Present*

- Recipient of the Chinese Government Scholarship (fully funded)
- Researching at the Key Laboratory of Pervasive Computing Lab supervised by Prof. Yuanchun Shi and Prof. Yuntao Wang

UNIVERSITAS INDONESIA Depok, Indonesia
Bachelor of Computer Science *Sep 2021 – Jan 2025*

- Graduated with Cum Laude honors, GPA: 3.88/4.00

WORK EXPERIENCE

TSINGHUA UNIVERSITY Beijing, China
Research Assistant *Sep 2025 – Present*

- Researching at the Key Laboratory of Pervasive Computing lab focusing on MLLMs, multimodal AI, human interaction, and intention reasoning with smart wearables.
- Contributing to EgoIntrospect, a multimodal egocentric dataset and benchmark for user-centric internal state reasoning; specifically responsible for the Proactive Request Recommendations task.

INTERNATIONAL OLYMPIAD OF ARTIFICIAL INTELLIGENCE (IOAI) Depok, Indonesia
Coach, Indonesian National AI Team *Jun 2025 – Aug 2025*

- Mentored and prepared Indonesia's top high school students for Indonesia's debut at the 2025 International Olympiad in Artificial Intelligence (IOAI), covering key areas such as machine learning, natural language processing (NLP), computer vision (CV), and data science.
- In Indonesia National Team first-ever participation in IOAI, we achieved an outstanding result: 3 Silver medals and 1 Bronze from our four delegations.

MAKNADATA Jakarta, Indonesia
AI Engineer Intern *Jan 2025 – Apr 2025*

- Worked closely with the CEO on the Criminal Detention Studies in Indonesia project, developing AI-driven solutions.
- Developing AI-driven solutions, including an LLM-powered system to extract key criminal information from unstructured legal documents (e.g., PDFs), an automated meeting notes generation system for prisoner-officer interviews, and a Chat RAG system leveraging relational databases for efficient information retrieval.

MEDIA KERNELS INDONESIA Jakarta, Indonesia
AI Engineer Intern *Sep 2024 – Dec 2024*

- Worked closely with the CTO to research, design, and develop an advanced Retrieval-Augmented Generation (RAG) system using vector databases to integrate diverse data sources, including social media, news, and legal documents.

TRAVELOKA Tangerang, Indonesia
Backend Engineer Intern *Feb 2024 – Jun 2024*

- Part of a core team within the company, the Flight Supply team, which is responsible for managing the technology related to airline inventory.
- Successfully improved booking health by $\sim$4%, seat selection availability of the top 3 airline providers by $\sim$10%, and our Memcache effectiveness and hit rate by $\sim$15%.

SABILAMALL Depok, Indonesia
Software Engineer Intern *Jun 2023 – Sep 2023*

- Revamped the backend of www.sabilamall.co.id using Express with TypeScript. This revamp resulted in an impressive average improvement of 50% in response time. Additionally, we took charge of managing and redesigning the MySQL database and successfully migrated it to AWS RDS.

XERATIC Tangerang, Indonesia
Data Ops Engineer Intern *Sep 2022 – Nov 2022*

- Contributed to the Great Giant Foods (GGF) project by utilizing Pentaho for ETL processes and Tableau to create informative dashboards.

FACULTY OF COMPUTER SCIENCE, UNIVERSITAS INDONESIA Depok, Indonesia
Teaching Assistant *Aug 2022 – Jan 2024*

- Teaching Assistant for Data Structure and Algorithm, Programming Foundations 1 & 2.
- Responsible for aiding professors and providing support to students in comprehending course materials. I also create lab exercises and programming tasks to enhance their learning experience.

ACHIEVEMENTS

1st Winner of Data Mining, GEMASTIK XVII | *Indonesian Ministry of Education* Sep 2024

- Achieved the gold medal in a prestigious national competition, outperforming 250+ teams through both our research, titled "Automated Assignment of Community Reports Using Early Fusion Multimodal Transformer," and our best performance in classifying tuition fee categories in the final round by leveraging cost-sensitive learning and a hill climbing ensemble method.

2nd Winner of Big Data Challenge, Satria Data 2024 | *Indonesian Ministry of Education* Aug 2024

- Achieved the silver medal by overcoming 400+ teams in a prestigious national competition by developing and implementing a unique social network analytical method for the study "Analysis of the 2024 Presidential Election Campaign Using Lexical Mutations Network, Public Stance, and Civility Tendency on Social Media."

1st Winner of Machine Learning Competition, Data Slayer | *Telkom Institute of Technology* Jan 2024

- Overcome other 120+ teams with proposed multi-layered stacking machine learning models to estimate the CO_2 vehicle emissions in Indonesia.

1st Winner of Data Competition, Airnology 2.0 | *Airlangga University* Oct 2023

- Created a highly effective "Image-based Drugs Search Engine using OCR and Levenshtein Algorithm". We also employed an ensemble method to predict rainfall during the selection phase.

2nd Winner of Natural Language Processing Data Competition | *Intelligo ID* Oct 2023

- Utilized NLP techniques to develop a "High-Accuracy, Low-Cost Model for Indonesian Logistics Sentiment Analysis".

Finalist of Data Mining, GEMASTIK XVI | *Indonesian Ministry of Education* Sep 2023

- Conducted a computer vision research project, utilizing CCTVs to create our own dataset. The objective of our study was to develop a paper titled "Helm Detection on Motorcycle Drivers in Indonesia using Deformable DETR".

Semi-Finalist of Big Data Challenge, Satria Data 2023 | *Indonesian Ministry of Education* Aug 2023

- Ranked in the top 10 by utilizing a fine-tuned model and Faster RCNN to develop an OCR for detecting Indonesian car plate numbers.

2nd Winner of Data Competition, JOINTS | *Gajah Mada University* May 2023

- Utilized an ensemble convolutional neural network architecture for detecting fires in images. Additionally, during the selection phase, we implemented an ensemble model for classifying earthquake damage.

2nd Winner of Datathon, Pekan RISTEK | *RISTEK FASILKOM UI* Dec 2022

- Utilized a multimodal model consisting of both images and text to assess whether the tweets contained any useful intelligence that could aid in disaster management and recovery efforts.

PUBLICATIONS

EgoIntrospect: An Egocentric Dataset and Benchmark for User-Centric Internal State Reasoning 2026

- Preprint: arXiv. Submitted to NeurIPS 2026 and currently under review.
- Introduced the first egocentric dataset captured in naturalistic, user-driven scenarios with self-annotations for AI assistant interaction; collected 180+ hours from 60 participants via a cross-device setup with synchronized video, audio, gaze, motion, physiological, and environmental signals.
- Formalized benchmark tasks for user-centric internal state reasoning across affective experience, interactive/request intent, and cognitive memory; specifically responsible for Task 2.2, Proactive Request Recommendations.

FrOG: Framework of Open GraphRAG 2025

- Published at the LLM-TEXT2KG @ ESWC 2025: Paper and Github.
- Conducted a study introducing an open GraphRAG, a system that integrates Retrieval-Augmented Generation (RAG) with knowledge graphs (KGs) for question-answering (QA). The research leveraged open source large language models (LLMs) to generate SPARQL queries on Wikidata, DBpedia and Local KGs.

Tutor Identity Classification using DiReC, a Two-Stage Disentangled Contrastive Representation 2025

- Achieved 3rd place overall in the BEA 2025 Shared Task @ ACL 2025, reaching a 0.9172 macro-F1 score: Paper and Github.
- Developed DiReC, a two-stage framework using disentangled representation learning and supervised contrastive learning to separate semantic content from stylistic features.

Evaluating State-of-the-Art Encoders for Multi-Label Emotion Detection 2025

- Co-developed an ensemble model for the 19th International Workshop on Semantic Evaluation (SemEval-2025) @ ACL 2025 that achieved a 56.58 average F1-macro score across all languages on the final leaderboard: Paper.
- Evaluated transformer fine-tuning versus classifier-only training for multilingual emotion classification across 28 languages using prompt-based encoders like mE5 and BGE.

Towards an Open NLI LLM-based System for KGs: A Case Study of Wikidata 2024

- Published in the International Seminar on Research of Information Technology and Intelligent Systems (ISRITI) 2024: IEEE Proceedings.
- Preliminary study for the **FrOG: Framework of Open GraphRAG** research. Developed an open GraphRAG-based QA system over Wikidata, leveraging LLM chaining to generate SPARQL queries.

Automatic Assignment of Community Reports on the CRM Platform Using Early Fusion Multimodal Transformer 2024

- Published in the Journal of Computer Sciences and Information (JIKI): Journal.
- Conducted an experiment that leveraged Transformer-based models such as DINOv2 and E5 to automatically assign community reports in Jakarta to the respective agencies, aiming to enhance the efficiency of the reporting process.

Helm Detection on Motorcycle Drivers in Indonesia using Deformable DETR 2024

- Published in the National Student Performance Bulletin in Information and Communication Technology: Journal.
- Conducted research that leveraged Transformer-based object detection model, Deformable DETR to detect helmet usage on motorcycle riders.

PROJECTS

What Does V-JEPA See? Sparse Autoencoders Reveal Interpretable Concepts in Video Foundation Models Jun 2026

- Developed a label-free interpretability pipeline that trains Top-K Sparse Autoencoders on frozen V-JEPA representations from Ego4D videos, then maps sparse features to real world concepts using narration mining, enrichment scoring, and Benjamini-Hochberg FDR correction across 141.9M tokens and 88 concepts. For more information: Slides.

MyoPong: Musculoskeletal Table Tennis Control with Training-Free High-Level Policy Dec 2025

- Developed MyoPong, a hierarchical framework for controlling musculoskeletal systems (up to 210 muscles) in table tennis, combining a physics-based high-level planner with a low-level PPO actor.
- Achieved a 94% success rate on the MyoChallenge phase 1 task, outperforming standard RL baselines through muscle synergy extraction and latent exploration (Lattice method). For more information: Website and Source Code.

Analysis of the 2024 Presidential Election Campaign Using Lexical Mutations Network, Public Stance, and Civility Tendency on Social Media Aug 2024

- Conducted research analytics project employed innovative methods like lexical mutation analysis, stance classification, and civility assessment to provide insights into the dynamics of social media discourse during the election campaign. For more information: Slides and Proposal.

PilgrimPal: Umrah AI Companion App Feb 2024

- Developed an app that contributes to helping pilgrims communicate, informing them of crowdedness in an area, and identifying their future needs. Our two main features are live CCTV crowd detection and an AI chatbot. For more information: Pitch deck and Prototype Release.

Duren Search Engine Dec 2023

- Developed a search engine website that utilizes a Learning to Rank technique. The technique is trained and indexed using the WikiCLIR dataset. For more information: Github Repository.

Image-based Drugs Search Engine using OCR and Levenshtein Algorithm Oct 2023

- Developed an OCR-based drug brand detector for the Kasir Pintar company during the Airnology 2.0 Data Competition. For more information: Slides and Github Repository.

High-Accuracy, Low-Cost Model for Indonesian Logistics Sentiment Analysis Oct 2023

- Conducted an experiment on a "High-Accuracy, Low-Cost" model for sentiment analysis using self-acquired and pseudo-labeled data obtained through web scraping. The focus of the analysis was on the Indonesian logistics industry. For more information: Slides and Proposal.

Segmentation Self-Driving Car Aug 2022

- This project involved training a semantic segmentation model using an image dataset from Cityscapes. Three experiments were conducted using three different models, namely U-Net, FCN8s, and U-Net x Mobile-Net V2. For more information: Github Repository or LinkedIn post.

VOLUNTEER EXPERIENCE

RISTEK FASILKOM UI Depok, Indonesia

Lead of Data Science & Analytics SIG | Lead of Web Development SIG *Feb 2024 – Jan 2025 | Jan 2023 – Jan 2024*

- Awarded as the best lead during the first quarter of RISTEK 2023 and second quarter of RISTEK 2024.
- Actively participated and contributed to achievement in various data science competitions.
- Collaborate on both internal and external projects with clients, while also taking charge of guiding and mentoring team members to enhance their proficiency in the data science and web development domain.

COMPFEST Depok, Indonesia

Data Science Academy Mentor *Aug 2024, Aug 2025*

- Guided and consulted a group of students on their final projects, providing advice on data science concepts, methodologies, and best practices.

ADDITIONAL INFORMATION

Programming Language Skills: Python, Java, TypeScript, Go, SQL, C++
AI Technical Skills: Computer Vision, NLP, LLM, VLM, Agentic System, PyTorch, Tensorflow, Hugging Face, LangChain, LangGraph, Weaviate, Milvus
Software Engineering Technical Skills: JavaScript frameworks, Go frameworks, Django, FastAPI, Springboot, Redis, Docker
Languages: Fluent in Indonesia (native) and English (IELTS: 7.5, Duolingo English Test: 145), Limited Proficiency in Chinese (HSK 4)
Certifications: TETRIS II: Data Analytics (2022); Indonesia AI (Machine Learning (2022), Deep Learning (2022), Computer Vision (2022)), AI Planet (Unsupervised Learning (2022), Natural Language Processing (2022)), CS50 Harvard: Intro to Computer Science (2021)